Cascaded $\chi^{(2)}$:$\chi^{(2)} \to \chi^{(3)}_{\text{eff}}$ in Singly-Resonant $LiNbO_3$ Metasurfaces

Sample Analysis (fabricated designs 1–4)

June 25, 2026

Abstract

A numerical companion to `Cascaded_chi3_SH_only_note.tex`, applying the singly-resonant (SH-only) cascaded-$\chi^{(3)}$ model to four fabricated $LiNbO_3$ metasurface designs supplied by the designer (samples 1–4, pump $\lambda_1 \approx 0.97$–$1.06\,\mu$m). Parameters are extracted from the COMSOL transmission spectra, the nonlinear overlap integrals, and the $LiNbO_3$ dispersion table; the full extraction and consistency checks are documented in `SH_only_sample_parameters.md`. We evaluate every observable that *survives* the loss of the fundamental resonance — the effective $\chi^{(3)}_{\text{casc}}(\Delta\omega)$ and its sign-flip, the two-photon-like loss, the bandwidth-integrated strength, the single-pass Kerr phase, saturable absorption, FWM, and SHG conversion — and locate the optimal operating detuning per sample. The headline is that the cascade is a *modest material enhancement* ($R^{(1\mathrm{R})} = |\chi^{(3)}_{\text{casc}}|/\chi^{(3)}_{\text{native}} \leq 2.5$, set by an overlap 2–10× below the value assumed in the note), that no single-pass Kerr *device* is observable, and that the highest-leverage design change is to raise the nonlinear overlap. All formulae are cross-checked against the note and an independent re-derivation (§10). Absolute magnitudes carry an $\mathcal{O}(1)$ normalisation-bridge uncertainty pending one absolute SHG-efficiency datum (§11).

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1 Samples and parameter extraction

The four samples share a 400 nm lattice period and a $LiNbO_3$ meta-atom (base H_1 + pillars H_2, total thickness H_t) on which only the second harmonic is resonant; the fundamental sits in the linear transmission band (measured FM transmission $|t|^2 \approx 0.73$, flat to $<1.3\%$, confirming the SH-only premise). The active tensor component is $\chi^{(2)}_{zzz} = 2d_{33}$ along the extraordinary axis (E_z out-of-plane), so $n_1 = n_e(\lambda_1)$, $n_2 = n_e(\lambda_2)$ from the supplied Sellmeier table. Frequencies satisfy $\omega_1 = \pi f_0(2\omega)$, $\gamma_2 = \omega_2/Q_2 = 2\pi\gamma[\mathrm{THz}]$, and the usable Kerr bandwidth $\Delta f = \gamma_2/2\pi = \gamma[\mathrm{THz}]$.

Table 1. Per-sample model parameters (files `design4_<geometry>.mat`). $\eta^{(1R)}_{geom}$ from the COMSOL overlap at $\Delta\omega = +2\gamma_2$ (baseline normalisation $C_{bridge} = 1$); material baseline $|\chi^{(2)}| = 5 \times 10^{-11}$ m/V, $\chi^{(3)}_{native} = 2 \times 10^{-21}\,\mathrm{m^2/V^2}$.

S	geometry	λ_1	λ_2	Q_2	Δf	$L_{meta} = H_t$	$\eta^{(1R)}_{geom}$
1	`alpha9`	997 nm	498 nm	3442	175 GHz	190 nm	0.0144
2	`H1_70`	975 nm	488 nm	3486	176 GHz	180 nm	0.0112
3	`H1_120_H2_100`	1058 nm	529 nm	4641	122 GHz	220 nm	0.0375
4	`H1_140_H2_80`	1064 nm	532 nm	5633	100 GHz	220 nm	0.0437

Overlap mapping. The designer's normalised overlap,

$$\mathrm{overlap} = \frac{1}{A}\,\frac{\iint E_z(\omega)\,E_z(\omega)\,E_z^*(2\omega)\,dA}{|E_z(\omega)|\,|E_z(\omega)|\,|E_z(2\omega)|}\,,$$

is exactly the SH-only SHG overlap $\widetilde{M}$ of the note (Eq. `Mtilde-def`), normalised. Since $\chi^{(3)}_{casc} \propto |\widetilde{M}|^2$, the geometric overlap is $\eta^{(1R)}_{geom}(\Delta\omega) = (|\mathrm{overlap}|/A)^2\,C_{bridge}$, evaluated at the SH frequency $f = f_0 + \Delta\omega/2\pi$. The resonant amplitude is divided out in the designer's definition, so the overlap is the *spatial coherence factor only*; the resonant build-up is supplied by the model's $1/(\Delta\omega + i\gamma_2/2)$. Reproducing $|\mathrm{overlap}|/A$ at $f_0 \pm 2\gamma$ from the raw arrays matches the designer's table to within $\pm 6\%$ (one tabulated value, sample 2 at -2γ, is a decimal typo: $0.9000 \to 0.0900$).

2 Cascaded $\chi^{(3)}$ spectra

With $\eta^{(1R)}_{geom}(\Delta\omega)$ from the overlap data, the effective susceptibility is

$$\chi^{(3)}_{casc}(\Delta\omega) \;=\; -\,\frac{\omega_1 |\chi^{(2)}|^2\,\eta^{(1R)}_{geom}(\Delta\omega)}{6\,n_2^2\,(\Delta\omega + i\gamma_2/2)}\,, \tag{1}$$

the singly-resonant boxed result of the note (Eq. `chi3-1R-explicit`).

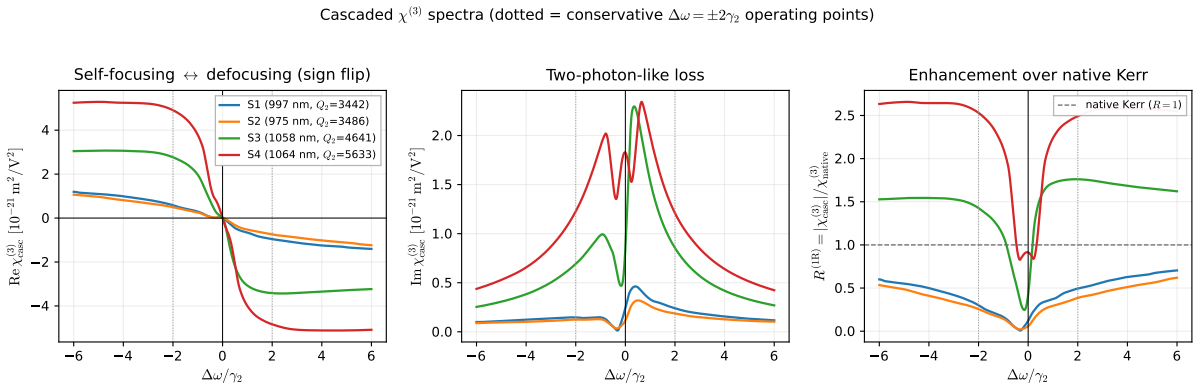


Figure 1. Cascaded $\chi^{(3)}$ versus SH detuning $\Delta\omega/\gamma_2$ for all four samples. (a) $\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}$ flips sign at $\Delta\omega = 0$ (self-focusing for $\Delta\omega < 0$, defocusing for $\Delta\omega > 0$). (b) $\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}$ (two-photon-like loss). (c) $R^{(1\mathrm{R})} = |\chi^{(3)}_{\mathrm{casc}}|/\chi^{(3)}_{\mathrm{native}}$; only S3, S4 clear the native-Kerr line. Dotted: conservative $\Delta\omega = \pm 2\gamma_2$.

Sign flip. $\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}} \propto -\Delta\omega$: a single metasurface delivers self-focusing, self-defocusing, or exact cancellation of the native Kerr by choice of $\Delta\omega$ (pump wavelength or post-fab SH trimming).

Loss peaks off resonance. Because the spatial overlap collapses at the exact SH resonance (a Fano feature in the COMSOL data), $\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}$ does *not* peak at $\Delta\omega = 0$ as the constant-η model predicts, but at finite detuning — a data-driven refinement of the note. The same effect makes $R^{(1\mathrm{R})}$ dip near $\Delta\omega = 0$ and rise into the wings (Fig. 1c, 3).

3 Overlap-limited enhancement and bandwidth

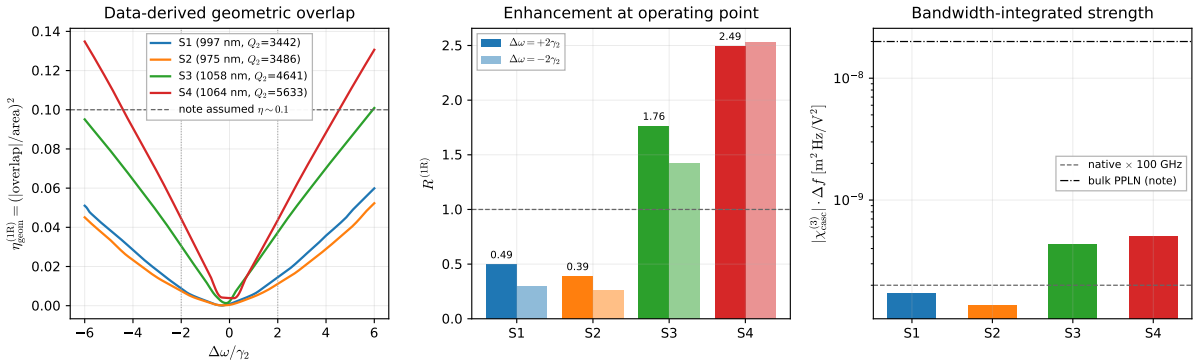


Figure 2. (a) Data-derived $\eta^{(1\mathrm{R})}_{\mathrm{geom}}(\Delta\omega) = (|\mathrm{overlap}|/A)^2$; all samples sit well below the note's assumed 0.1 at $\pm 2\gamma_2$ and collapse at $\Delta\omega = 0$. (b) $R^{(1\mathrm{R})}$ at $\Delta\omega = \pm 2\gamma_2$. (c) Bandwidth-integrated strength $|\chi^{(3)}_{\mathrm{casc}}| \cdot \Delta f$, versus native×100 GHz and bulk PPLN.

Ranking. $R^{(1\mathrm{R})}$ = S4 2.49 > S3 1.76 > S1 0.49 > S2 0.39. Only samples 3 and 4 exceed native $LiNbO_3$ Kerr. The binding constraint is the overlap: $\eta^{(1\mathrm{R})}_{\mathrm{geom}} = 0.011$–$0.044$, a factor 2–10 below the ~ 0.1 assumed in the note (§9, R1).

Bandwidth-integrated strength. Even the best sample sits $\sim$ 40–100× below bulk PPLN in $|\chi^{(3)}_{\mathrm{casc}}| \cdot \Delta f$. The metasurface case is chip-scale form factor and CW-threshold operation, not raw integrated nonlinear strength.

4 Optimal detuning

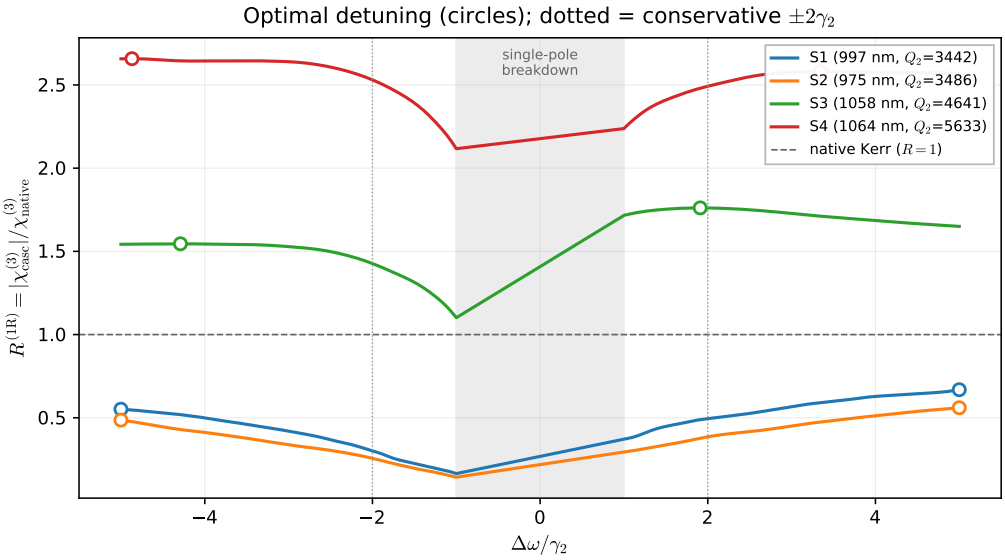


Figure 3. $R^{(1R)}(\Delta\omega)$ with the per-branch optima (circles) and the conservative $\pm 2\gamma_2$ points (dotted). R dips inside the $|\Delta\omega| < \gamma_2$ single-pole breakdown band (shaded) and forms a broad plateau in the wings.

Scanning $\Delta\omega$ over the validity window $|\Delta\omega| \in [1, 5]\gamma_2$ with the data-derived $\eta^{(1R)}_{\mathrm{geom}}(\Delta\omega)$:

Table 2. Optimal detuning per sample (best $|\chi^{(3)}_{\mathrm{casc}}|$ on each branch) versus the conservative $+2\gamma_2$ point.

S	best defocus ($\Delta\omega > 0$)	best focus ($\Delta\omega < 0$)	gain over $+2\gamma_2$
1	$+5\gamma_2$, R 0.67	$-5\gamma_2$, R 0.55	$\times$1.35 (still < 1)
2	$+5\gamma_2$, R 0.56	$-5\gamma_2$, R 0.49	$\times$1.46 (still < 1)
3	$+1.9\gamma_2$, R 1.76	$-4.2\gamma_2$, R 1.55	$\times$1.00
4	$+3.2\gamma_2$, R 2.58	$-4.8\gamma_2$, R 2.66	$\times$1.04

Takeaway. Detuning is *not* the lever. For the two useful samples the conservative $\Delta\omega = 2\gamma_2$ is already within $\sim$5% of the achievable optimum; operating closer ($|\Delta\omega| < \gamma_2$) loses, because the overlap collapses there and the single-pole/adiabatic model breaks. The weak samples cannot be detuned past native Kerr at any point in the trusted window.

5 Feasibility: single-pass phase and saturable absorption

The single-pass per-photon Kerr phase and intensity-dependent loss are (note Eqs. `phi-1R`, `Im-chi3-1R`)

$$\varphi^{(1R)}_{\mathrm{NL}} = \frac{3\omega_1}{2c\,n_1}\,\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}\,|E_{\mathrm{loc}}|^2 L_{\mathrm{meta}}\,, \qquad \left|\frac{\Delta T}{T}\right| = \frac{3\omega_1}{c\,n_1}\,\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}\,|E_{\mathrm{loc}}|^2 L_{\mathrm{meta}}\,, \tag{2}$$

with $|E_{\mathrm{loc}}|^2 = 2T^2_{\mathrm{geom}} I_{\mathrm{in}}/(\varepsilon_0 c\, n_1^2)$ and $T_{\mathrm{geom}} = 1$ (baseline).

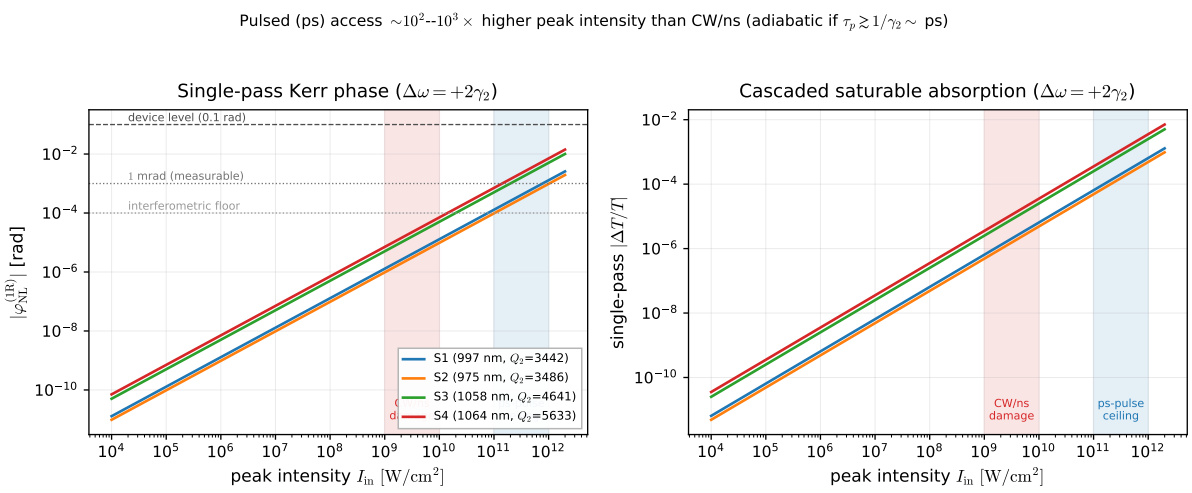


Figure 4. (a) Single-pass Kerr phase $|\varphi_{\rm NL}|$ and (b) saturable absorption $|\Delta T/T|$ vs. peak intensity at $\Delta\omega = +2\gamma_2$. Shaded: the CW/ns damage band (1–10 GW/cm^2, red) and the ps-pulse fluence ceiling ($\sim$0.1–1 TW/cm^2, blue); the device level (0.1 rad), 1 mrad, and the interferometric floor are marked.

CW: nothing observable. At the CW/ns damage threshold the best sample reaches only $\varphi_{\rm NL} \sim 10^{-4}$ rad, and CW operation is anyway capped far lower by green-SH photorefractive damage. No CW Kerr nonlinearity is observable; removing the FM resonance removes the Q_1^2 build-up, as the note predicted.

Pulsed (ps): the $\chi^{(3)}$ becomes measurable. Pulsed pumping helps substantially. It accesses 10^2–$10^3\times$ higher peak intensity *and* sidesteps the CW photorefractive limit, lifting the phase to $\sim$mrad and $\Delta T/T$ to $\sim 10^{-3}$ — enough to *measure* the $\chi^{(3)}$, though still $\ll 0.1$ rad (no device). Power cannot recover the lost Q_1^2. These estimates are quasi-static; since the experimental $\tau_p \sim$ 1–3 ps is comparable to the SH lifetime we test them with a time-domain pulse simulation (§6).

6 Time-domain pulse simulation

Since $\tau_p \sim$ 1–3 ps is comparable to the SH lifetime $1/\gamma_2 =$ 0.9–1.6 ps, we integrate the actual SH-only TCMT (note Eq. `a2-1R`) with the FM responding instantaneously, so the drive follows the pulse intensity envelope:

$$\dot{a}_2 = (i\Delta\omega - \tfrac{\gamma_2}{2})\, a_2 + i\,\beta(t), \qquad \beta(t) \propto |E_{\rm loc}(t)|^2 \propto I(t). \tag{3}$$

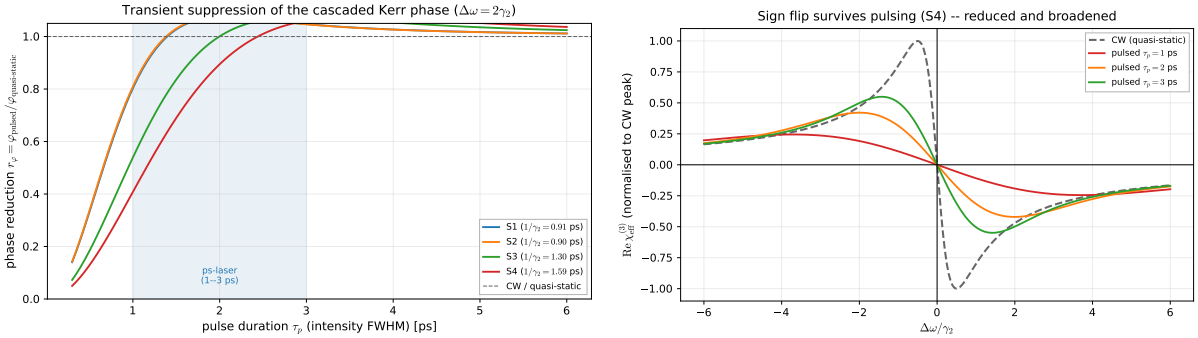


Figure 5. Left: nonlinear-phase reduction $r_\varphi = \varphi_{\rm pulsed}/\varphi_{\rm quasi\text{-}static}$ vs pulse duration (1–3 ps experimental band shaded). Right: the cascaded $\mathrm{Re}\,\chi^{(3)}_{\rm casc}$ sign flip survives pulsing (reduced and spectrally broadened, still flips at $\Delta\omega = 0$).

The right observable. The *energy*-integrated nonlinear phase is conserved for any pulse length (a sum rule — the SH delivers its full action, just spread over $\sim 1/\gamma_2$). The reduced, Z-scan-relevant quantity is the I^2-weighted / peak phase.

Result. At $\Delta\omega = 2\gamma_2$ the phase reduction is $r_\varphi(\tau_p = 1\,\mathrm{ps}) = 0.80$ (S1/S2), 0.54 (S3), **0.41** (S4), rising to $r_\varphi \approx 0.9$–1.07 for $\tau_p \geq 2$ ps (slight resonant overshoot). **The quasi-static estimate holds for $\tau_p \gtrsim 2$ ps**; the penalty bites only at 1 ps on the highest-Q samples ($\sim 2.5\times$ on S4). **Use $\tau_p \gtrsim 2$ ps.** The sign flip survives, so the sign-flip experiment is robust at ps durations.

7 FWM and SHG conversion

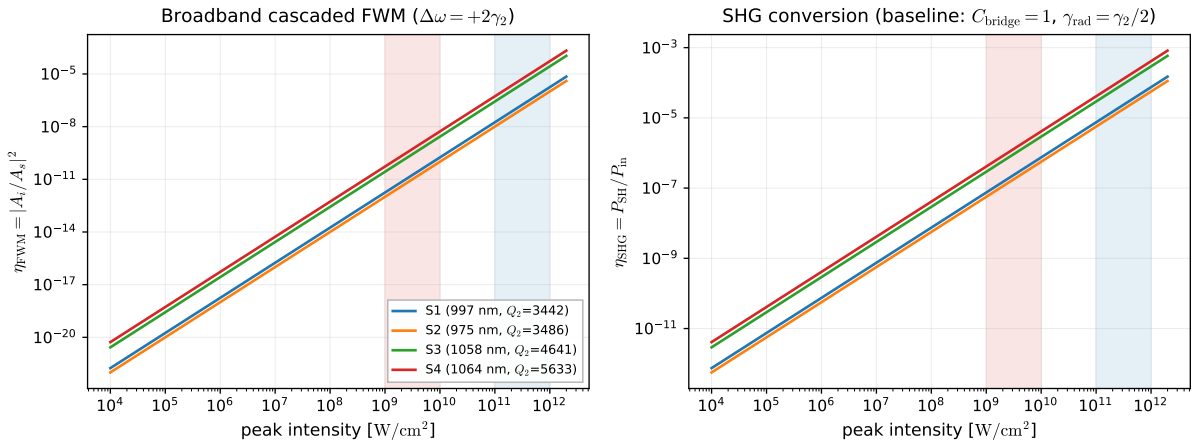


Figure 6. (a) Broadband (phase-matching-free) cascaded FWM efficiency $\eta_{\rm FWM} = |A_i/A_s|^2$ and (b) SHG conversion $\eta_{\rm SHG} = P_{\rm SH}/P_{\rm in}$ versus intensity at $\Delta\omega = +2\gamma_2$. SHG is an order-of-magnitude baseline ($C_{\rm bridge} = 1$, $\gamma_{\rm rad} = \gamma_2/2$).

FWM. Efficiency reaches $\sim 10^{-4}$ only at the damage threshold; the sub-wavelength thickness makes it phase-matching-free (full γ_2 bandwidth), useful as a *diagnostic* of $\chi^{(3)}_{\rm casc}$ rather than as a device.

SHG — calibration only, not a target. SHG is a $\chi^{(2)}$ process and *not* an objective of this $\chi^{(3)}$ study. We use it only because it shares the same overlap and SH resonance: measuring $\eta_{\rm SHG}$ ($\sim 10^{-7}$ at 1 GW/cm^2 baseline) vs. detuning and power *pins the absolute* $\chi^{(3)}_{casc}$ *scale* (the normalisation bridge, §11) and validates the cascade's first ($\chi^{(2)}$) step.

Measuring the sign flip (the $\chi^{(3)}$ observable). Pump at ω_1 (always); scan the pump *wavelength* so its *generated* harmonic $2\omega_1$ crosses the SH resonance ω_2 ($\Delta\omega = 2\omega_1 - \omega_2 : - \to +$). A closed-aperture Z-scan at each wavelength reads the sign of n_2 from the trace ordering — valley→peak = self-focusing ($n_2 > 0$), peak→valley = defocusing; the cascade is **odd in** $\Delta\omega$ and flips at $\Delta\omega = 0$ (Fig. 1a). Native $LiNbO_3$ Kerr is a constant positive background, so a *full* reversal of the total n_2 needs $R > 1$ (S3/S4); regardless, isolate the cascade as the **antisymmetric-in-$\Delta\omega$** component. Only ~ 1 nm of pump tuning spans $\pm 3\gamma_2$ at $Q_2 \approx 5000$. Equivalent probes: SPM/XPM spectral blue/red asymmetry, or spectral interferometry for the signed phase.

8 Observable verdict

Table 3. Which effects are reasonable to observe with these samples.

Effect	Status in SH-only	Observable here?
Sign flip (focus↔defocus)	$\chi^{(3)}$, dispersive shape	**Yes, qualitatively** (scale-robust)
$\chi^{(3)}_{\mathrm{casc}}$ / R (material)	material property	Indirect; S3/S4 only (R 1.8–2.5)
Per-photon Kerr phase φ_{NL}	single-pass	Only pulsed, ∼mrad; $\ll 0.1$ rad
Saturable absorption $\Delta T/T$	single-pass Im $\chi^{(3)}$	Only pulsed ($\sim 10^{-3}$)
Broadband FWM	single-pass	Pulsed diagnostic
SHG conversion efficiency	$\chi^{(2)}$ (out of scope)	Calibration of $\chi^{(3)}_{\mathrm{casc}}$ scale only
Bistability, switching, limiter, AM modulator	*die* (no FM cavity)	No, at any power

9 Design rules for the optimal device

R1. Maximise the nonlinear overlap (dominant, linear lever). $\chi^{(3)}_{\mathrm{casc}} \propto \eta^{(1\mathrm{R})}_{\mathrm{geom}} = |\mathrm{overlap}|^2$. The samples sit at $|\mathrm{overlap}| \approx 0.10$–$0.21$; reaching $R = 4$ at S4's Q needs $|\mathrm{overlap}| \approx 0.26$, $R = 10$ needs ≈ 0.42. Target $|\mathrm{overlap}| \gtrsim 0.3$ at the operating detuning via stronger qBIC asymmetry (staying absorption-limited, where $|\mathrm{overlap}|^2 Q$ grows with asymmetry), sub-domain (unit-cell) poling of $\chi^{(2)}$, or pump-near-field shaping.

R2. Fix the detuning at $\Delta\omega \approx 2\gamma_2$; don't chase it (§4): within $\sim 5\%$ of optimal, TPA penalty $(\gamma_2/2\Delta\omega)^2 = 1/16$, adiabatic approximation clean.

R3. Push Q_2 with diminishing returns and a disorder floor. $\chi^{(3)}_{\mathrm{casc}} \propto Q_2$, but only one power of Q helps in 1R, bandwidth $\propto 1/Q_2$, and the array linewidth must beat disorder ($\sigma_\omega/\omega \lesssim 1/Q_2 \approx 2 \times 10^{-4}$ at $Q_2 = 5000$). Verify the *array* (not single-element) Q.

R4. Thicker $LiNbO_3$. Single-pass φ_{NL}, $\Delta T/T$, FWM all $\propto L_{\mathrm{meta}}$ (not $\chi^{(3)}_{\mathrm{casc}}$ itself).

R5. Pump with ps pulses; use MgO-doped $LiNbO_3$. The SH is green (488–532 nm), where CW is photorefractively limited; ps pulses ($\tau_p \gtrsim 1/\gamma_2 \sim$ps) give 10^2–$10^3\times$ more peak intensity and escape that limit, lifting the $\chi^{(3)}$ signals into the measurable range (∼mrad).

R6. Confirm Q is the loaded value. An FWHM read of the SH dip gives $\approx 0.6\times$ the quoted Q; the model needs the total γ_2 and $\chi^{(3)}_{\mathrm{casc}} \propto Q_2$.

R7. For a CW Kerr *device*, go doubly-resonant. Even fully overlap-optimised, the SH-only per-photon phase lacks the Q_1^2 FM-cavity build-up and stays μrad-class. SH-only's

best roles are a $\chi^{(3)}$ *diagnostic* (sign flip + FWM; SHG only to calibrate the scale) and *materials-level* $\chi^{(3)}$; a bistability/switching device needs the FM qBIC added back.

R8. Match sample to pump ($\lambda_1 = 975$–$1064\,\mathrm{nm}$; S4 = Nd:YAG) and trim the SH resonance post-fabrication to set $\Delta\omega$.

10 Verification

All formulae were re-derived from the note and cross-checked two independent ways (`cascaded_chi3/sample_optimal.py` and an independent reviewer): the prefactor-free identity $\mathrm{Re/Im}\,\chi^{(3)}_{\mathrm{casc}} = -2(\Delta\omega/\gamma_2)$ holds exactly; the lineshape matches the dimensionless adiabatic forms of `analytic.py` to 2.5×10^{-16}; units reduce to $\mathrm{m^2/V^2}$ ($\chi^{(3)}_{\mathrm{casc}}$) and rad ($\varphi_{\mathrm{NL}}$); and the sample-4 spot value $|\chi^{(3)}_{\mathrm{casc}}| = 4.98 \times 10^{-21}$, $R = 2.49$ was reproduced by hand. The Boyd degeneracy factors (3 for SPM/phase, 2× for loss-vs-phase, cubic pump in FWM), the ε_0 cancellation, and the squared overlap in $\eta^{(1\mathrm{R})}_{\mathrm{geom}}$ are all correct. **No discrepancies.**

11 Open items and the absolute scale

Rankings, lineshapes, sign-flip, and Q/η scalings are robust. Absolute magnitudes ($\chi^{(3)}_{\mathrm{casc}}$, φ_{NL}, $\Delta T/T$, η_{FWM}, η_{SHG}) scale with the normalisation bridge C_{bridge} (the 2-D-overlap → 3-D-Brillouin mapping, baseline 1), with $|\chi^{(2)}|^2$, and with $1/\chi^{(3)}_{\mathrm{native}}$ — all $\mathcal{O}(1)$ at present. A single absolute SHG-efficiency datum (one full nonlinear simulation) pins the bridge; the full open-question list is in `SH_only_designer_questions.md`. Other open items: the undocumented `symmetry` field, the loaded-vs-intrinsic Q, the radiative/absorptive split, and refined $d_{33}/\chi^{(3)}_{\mathrm{native}}$ at $\sim 1\,\mu\mathrm{m}$.

12 Reproduce

Run `conda run -n ml_base python cascaded_chi3/<script>.py` with `<script>` one of:

- `sample_params` — parameters + consistency checks;
- `sample_effects` — effects + figures;
- `sample_optimal` — cross-checks + optimal detuning;
- `sample_pulsed` — time-domain pulsed TCMT.

Figures are written to `cascaded_chi3/figures/samples/`. Companion markdown: `SH_only_sample_parameters.md` (extraction), `SH_only_sample_analysis.md` (analysis), `SH_only_designer_questions.md` (designer queries).